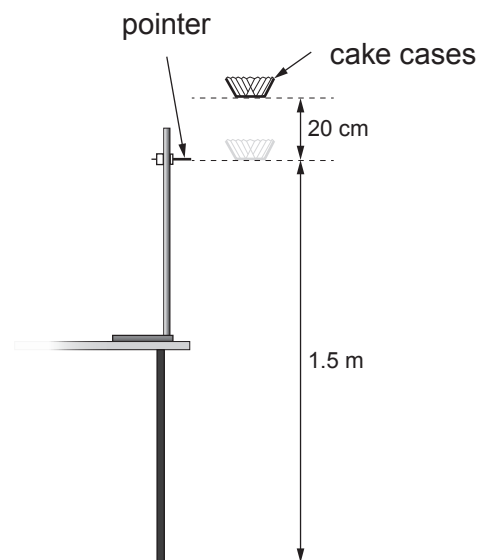


8. A class carries out an investigation into the relationship between the terminal speed of paper cake cases and their mass. They let the cake cases drop from about 20 cm above a pointer which is 1.5 m above the floor. They time how long they take to drop from the pointer to the floor. They assume that after 20 cm the cake cases will be travelling at terminal speed.



- (a) Explain, in terms of two named forces, why the cake cases travel at terminal speed. [2]

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- (b) The students take some trial readings to help them determine the number of repeat readings they need to take. Here are their results for 1 cake case.

Mass (g)	Time for cake case to travel 1.5 m (s)							
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8
0.5	1.16	1.19	1.17	1.29	0.72	1.22	1.24	1.15

- (i) **Circle** the anomalous result. [1]

- (ii) Calculate the mean time. [2]

Mean time = s

- (iii) Use an equation from page 2 to calculate the mean speed. [1]

Mean speed = m/s

- (c) The students then carry out their experiment. First they measure a height of 1.5 m with two metre rulers and set their pointer. They drop 1 cake case and record the time taken to drop using a stopwatch. This is repeated 5 times. They then repeat the experiment with 2, 3, 4 and 5 cake cases in a stack to vary the mass of the cake cases.

- (i) State the independent variable in this experiment. [1]

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- (ii) State **one** controlled variable in this experiment. [1]

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- (iii) Explain how the data could be measured more accurately. [2]

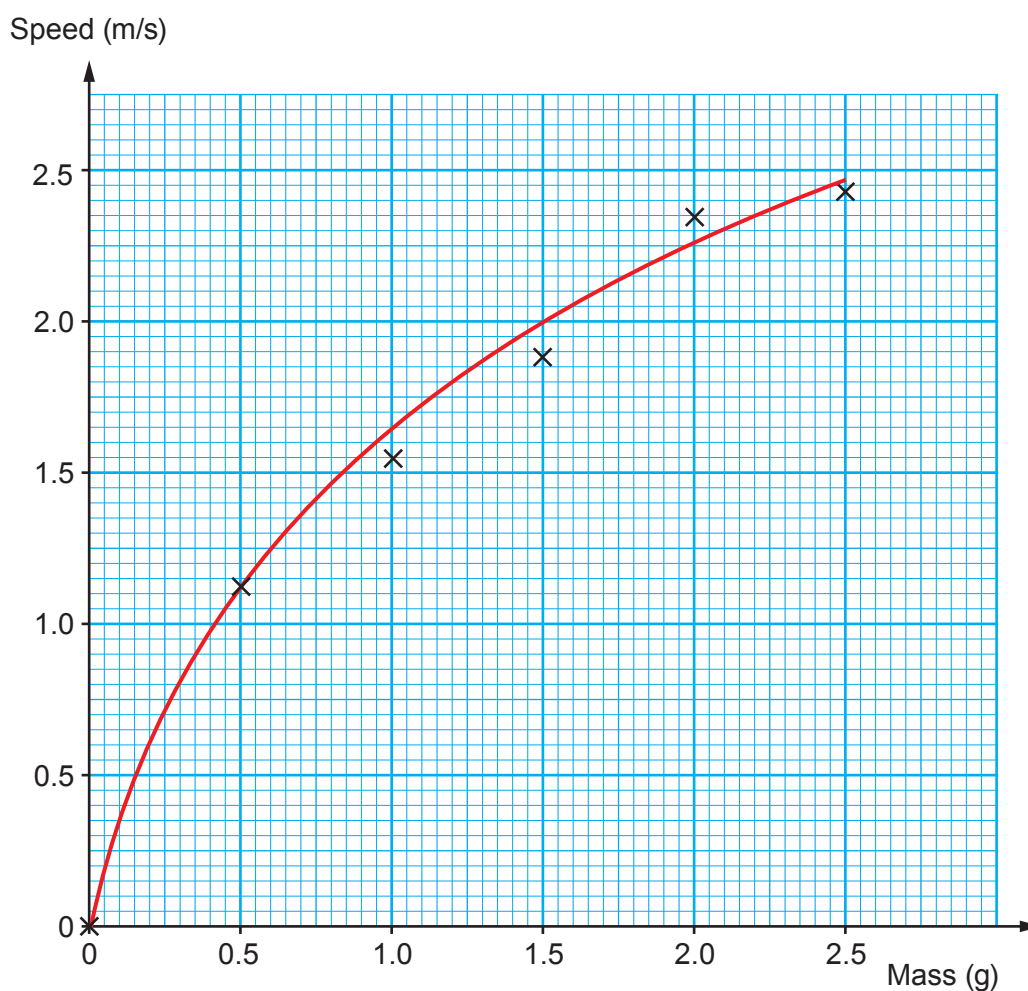
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(d) The results from one group are shown on the graph below.



Angus concludes that when the mass doubles, the speed is always 1.5 times bigger. Explain whether the results support his conclusion. Use data from the graph to support your answer. [3]

Space for calculations.

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